Haocheng He

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PROFESSIONAL SUMMARY

Industrial Engineering graduate with a Law minor from GDUT, currently pursuing an MSc in Aviation Engineering and Operations Management at PolyU (GPA: 3.95/4.3). Experience spans airport emergency dispatch simulation, manufacturing line optimization, and AI hardware integration, with 3 published invention patent applications.

EDUCATION

The Hong Kong Polytechnic University — Hong Kong SAR, China
Master of Science in Aviation Engineering and Operations Management — Sep 2025 – Jan 2027
GPA: 3.95 / 4.3 (Current)
Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty

Guangdong University of Technology — Guangzhou, China
Bachelor of Engineering in Industrial Engineering — Sep 2021 – Jun 2025
GPA: 3.67 (Top 20%)
Thesis: AnyLogic-Based Simulation Optimization of Emergency Vehicle Route Planning in Large Airports
Honors: Excellent Graduation Thesis + Innovation Award

Bachelor of Laws (Minor Degree in Law) — Sep 2022 – Jun 2025
GPA: 3.61
Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management

EXPERIENCE

AI Engineer Assistant — May 2025 – Jun 2026
Shenzhen Wuxian Hongyin Technology Co., Ltd. — *Shenzhen, China*
- Supported module integration for an Ascend-based piano fingering recognition system (Python/C++); reviewed architecture and helped locate issues.
- Maintained Ubuntu servers and Huawei switches/routers; also supported procurement, administration, and technical interviews.

Research Assistant — Sep 2023 – Jun 2025
Guangdong University of Technology (with Guangzhou Baiyun Int'l Airport) — *Guangzhou, China*
- Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios.
- Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D* Lite, A*, Dijkstra, BFS, and GA routing algorithms.
- Completed 4 emergency-scenario comparisons across 5 routing algorithms; D* Lite achieved the highest average reliability index (1.342).

Industrial Engineering Intern — Apr 2024 – May 2024
Midea Group — Kitchen & Water Heater Division — *Foshan, China*
- Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps.
- Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%, reducing headcount from 20 to 18.

Manufacturing Operations Intern — Jul 2023 – Aug 2023
Guangxi LiuGong Machinery Co., Ltd. — *Liuzhou, China*
- Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.
- Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation. Submitted recommendations to the production-line team for evaluation.

PROJECTS

Aircraft Engine Remaining Useful Life Prediction (Course Project) — Apr 2024 – Jul 2024
Guangdong University of Technology — *Guangzhou*
- Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA.
- Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.

SKILLS & INTELLECTUAL PROPERTY

Programming: Python, Java, SQL/MySQL, C/STM32, JS/TS **ML:** LSTM, BiLSTM, Transformer, RF, PCA **AI:** LLM Workflows, Prompt Eng., Loop Engineering
Simulation: AnyLogic, Plant Sim., OR-Tools, Dijkstra, A*, BFS, GA, ECRS **Tools:** Ubuntu/Linux, Git, JLC EDA, Oscilloscope
Languages: English (IELTS 6.5), Mandarin (native), Cantonese (conversational)

Patents: *MCMC Sampling-Based Motion Control Method for a PI Coating Machine Controller* (CN122085794A, 2026); *Coating Control Method and System Based on Deep Reinforcement Learning* (CN121325552A, 2025); *Energy Regulation Method and System for a Household Solar Energy Storage System* (CN117767437A, 2024). Inventors include Xianzhong Zhou, Haocheng He et al.
Utility Model: *Integrated Equipment for Wire Harness Production* (CN221352455U, 2024). **Software Copyrights:** Financial Data Analysis V1.0 (2024SR0146827); Flood Emergency Dispatch V1.0 (2024SR0549247).